

technique of serving all the processors with their requested data within one cycle. Hence the processors can go on computing huge amounts of data without having to wait their turn to get more data.

In addition, graphics cards or GPUs come with 'texture memory' which is memory specifically set aside to ensure faster read-write operations and faster computations for 3D renderings. Since the GPU is designed to process 3D calculations fast and efficiently, it will have local access to the texture images, allowing it to complete a scene much faster than having to use memory stored in another physical location within the computer. Such an implementation would obviously be very beneficial in critical conditions where time is of utmost importance.

Just imagine a failed Curiosity Mars rover landing due to slow radar readings and estimations. Had that happened, it would have set the U.S. back by more than \$2 billion and cost a delay of more than a couple of years to investigate the Red Planet. RAM speed in embedded systems in the context of an interplanetary project is especially very important. Every single detail can make a huge impact in such situations where milliseconds matter and there's no room for error.

Displays – Challenges and considerations

In many cases, an embedded system doesn't need displays and so displays have no relevance in such products. In products that do require visual indicators, a small panel of LED lights usually suffice - e.g. the audio system in your car or the control panel on a microwave oven. There are other applications of embedded systems for which CRT-type displays (medical equipment and electronic frequency readers) are used.

The type of display required will determine whether you'll face difficulties in designing this component. For systems with minimal visual indicators, the circuit required to control the display is controlled by a chip, and this can also be done directly by the processor.

However, certain systems may need extensive use of displays. The best example is your mobile phone. Yes, this would include every phone from the most basic to the likes of BlackBerrys and iPhones. If you care about the details of a smartphone's specs, you might have observed that most Android phones (since they run on a wider variety of hardware than other mobile OSes) are built around a specification predefined by a SoC (System-on-Chip) design company which requires details about the processing power, memory included, memory handling capabilities, communication ports available, on-board graphics processors and maximum supported resolution of the display to be mentioned. While a